

Distributed Systems Fundamentals

Distributed systems are computing systems whose components are located on different networked computers, which communicate and coordinate their actions by passing messages to one another from any system.

Key Concepts: 1. Scalability: The ability to handle increasing amounts of work by adding resources. 2. Reliability: System continues to function even if some components fail. 3. Performance: Response time and throughput meet requirements. 4. Maintainability: System is easy to modify and improve.

Load Balancing in Distributed Systems: Load balancing is a technique to distribute workload evenly across multiple computing resources. It improves resource utilization, maximizes throughput, minimizes response time, and avoids overload on individual resources. Common strategies include Round Robin, Least Connections, and Weighted Distribution.

Fault Tolerance: Fault tolerance is the ability of a system to continue operating properly in the event of the failure of some of its components. In distributed systems, this is achieved through redundancy, replication, and recovery mechanisms.

Distributed Databases: Distributed databases span multiple physical locations and are replicated across multiple sites. They provide better performance through parallelism, improved reliability through redundancy, and increased availability by having data stored at multiple sites.